UNIT - V

QUALITY SYSTEMS

ISO 9000 - Implementation - Documentation - Auditing - QS 9000 - ISO 14000 - Case Studies

1. Need for ISO 9000 and Other Quality Systems

- A **quality system** (Quality Management System - QMS) = the organisational structure, responsibilities, procedures, processes and resources for implementing quality management.
- **ISO** = International Organisation for Standardisation, Geneva (1947). "ISO" comes from the Greek word *isos* = equal. **ISO 9000 series first issued in 1987** (based on British Standard BS 5750 and military standards); revised 1994, 2000, 2008, **2015.**

Why do we need ISO 9000?

- **Customer confidence:** assurance that the supplier has a documented and audited system; no need for customer to audit each supplier (second party audit) - one certificate accepted everywhere.
- **Global market access:** international trade, tender requirement, preferred supplier.
- Consistent quality of product and service; fewer defects and customer complaints.
- Better internal management: clear responsibilities, documented procedures, training.
- Reduces cost of quality, waste and rework; improves productivity.
- Legal and statutory compliance, product liability protection.
- Competitive advantage and improved image and employee morale.
- Starting point and base for TQM (system + continual improvement).

The ISO 9000 family

Standard	Purpose
ISO 9000	Fundamentals and vocabulary
ISO 9001	**Requirements** of QMS (the standard used for certification)
ISO 9004	Guidelines for performance improvement / sustained success
ISO 19011	Guidelines for auditing management systems

- Earlier (1994 version): ISO 9001 (design, production, installation, servicing), 9002 (no design), 9003 (final inspection). In 2000 all merged into **ISO 9001:2000.**

Other quality systems

- **QS 9000** (automotive - Chrysler, Ford, GM), **ISO/TS 16949 -> IATF 16949,** AS 9100 (aerospace), TL 9000 (telecom), ISO 13485 (medical devices), ISO 22000 (food safety), **ISO 14001** (environment), OHSAS 18001 -> **ISO 45001** (health and safety), ISO/IEC 27001 (information security), ISO 17025 (labs), **ISO 26000** (social responsibility).

> Remember: certification is given by an independent **certification body** (registrar); the standard is not a product standard - it is a system standard.

2. ISO 9000:2000 Quality System - Elements

Eight quality management principles (ISO 9000:2000)

1. **Customer focus**
2. **Leadership**
3. **Involvement of people**
4. **Process approach**
5. **System approach to management**
6. **Continual improvement**
7. **Factual approach to decision making**

8. Mutually beneficial supplier relationships



Process model of ISO 9001:2000

- Customers' requirements (input) -> **Management responsibility** -> **Resource management** -> **Product realisation** -> **Measurement, analysis and improvement** -> product -> customer satisfaction (output).
- With the PDCA cycle applied to all processes; feedback of customer satisfaction.

Clauses of ISO 9001:2000 (sections 4 to 8)

Clause	Title	Main requirements
4	Quality Management System	General requirements, documentation: quality manual, control of documents and records
5	Management responsibility	Management commitment, customer focus, quality policy, planning and objectives, responsibility and authority, management representative, management review
6	Resource management	Provision of resources, human resources (competence, training), infrastructure, work environment
7	Product realisation	Planning, customer related processes, design and development, purchasing, production and service provision, control of monitoring and measuring devices

8	Measurement, analysis and improvement	Customer satisfaction, internal audit, monitoring of process and product, control of nonconforming product, data analysis, continual improvement, corrective and preventive action

- (Clauses 1 to 3: scope, normative reference, terms and definitions.)

Key elements explained

- **Quality policy and objectives:** measurable, consistent and communicated.
- **Management review:** top management reviews the QMS at planned intervals - inputs: audit results, customer feedback, process performance, status of corrective actions; outputs: improvement decisions and resource needs.
- **Design control:** planning, inputs, outputs, review, verification, validation, changes.
- **Purchasing:** evaluation and selection of suppliers, verification of purchased product.
- **Control of nonconforming product:** identify, segregate, disposition (rework, scrap, concession).
- **Corrective action** (remove the cause of an existing nonconformity) and **preventive action** (remove cause of a potential one).
- **Calibration** of measuring equipment traceable to national standards.
- **Training, identification and traceability, handling, storage, packaging, delivery.**

ISO 9001:2015 (new) - quick note

- High level structure (Annex SL) with 10 clauses: Scope, References, Terms, **Context of the organisation,** Leadership, Planning **(risk based thinking),** Support, Operation, Performance evaluation,

Improvement.

- No mandatory quality manual; "documented information"; 7 principles (the supplier principle becomes "relationship management", the system approach is merged with process approach, "evidence based decisions").

3. Implementation of Quality System



Steps for implementing ISO 9001

1. **Top management commitment** and decision; appoint the **management representative (MR).**
2. Form the **steering committee** and project team; make an implementation plan with timeline.
3. **Awareness and training** of all employees (basics of ISO, internal auditor training).
4. **Gap analysis / initial review:** compare existing practice with the standard; identify the gaps.
5. **Document the system:** quality manual, procedures, work instructions, formats; define processes.
6. **Implement** the system - practice what is documented; maintain records; train as per the documents.
7. **Internal audits** by trained auditors; corrective actions; **management review.**
8. Choose and apply to an accredited **certification body; pre-assessment** (optional).
9. **Stage 1 (document review) and Stage 2 (on-site) audit** by the registrar.

10. Close non-conformities -> **certificate** (valid 3 years); **surveillance audits** every 6 or 12 months; recertification after 3 years.
- Time taken: 6 to 12 months; cost: consultant, training, certification fees.

Success factors
- Real commitment (not just a certificate), involvement of employees, simple documents (do not over-document), proper training, use of consultants only as guides, regular audits and review, link with business goals.

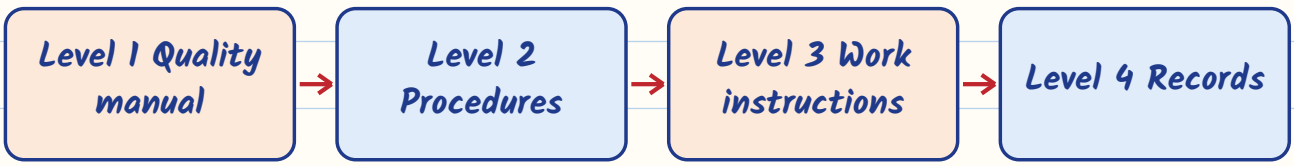
Common pitfalls
- Treating ISO as paperwork; copying other's manual; lack of ownership; poor internal audits; certificate and then relaxation.

Benefits of implementation
- Improved processes and customer satisfaction, reduced rejections, system discipline, better supplier control, marketing advantage and global acceptability.

4. Documentation

> Documentation provides **evidence** that the system is defined, understood and implemented - **"Say what you do, do what you say, prove it"** (record it).

Four levels (pyramid) of documents



- **Level 1 - Quality Manual:** describes the QMS - scope, quality policy and objectives, organisation chart, summary of procedures, process interaction. (Short, 20-40 pages; policy for "what and why".)

- **Level 2 – Procedures (SOPs): who, what, when, where** – the way activities are carried out across departments (e.g., purchase, internal audit, control of nonconforming product, corrective action, control of documents, control of records – the six mandatory procedures of ISO 9001:2000).
- **Level 3 – Work instructions, specifications, drawings, standards: how** the task is done at the work place (step-by-step; check lists).
- **Level 4 – Records, formats, forms, reports: evidence** of the activities and results – inspection reports, audit reports, calibration records, training records, minutes of management review.

Control of documents

- Approval before issue; review and update; identification of revision status; availability at points of use; legible; **obsolete documents** removed or marked; external origin documents controlled; distribution list.

Control of records

- Identification, storage, protection, retrieval, **retention time** and disposition.

Good documentation tips

- Simple, clear language; use flowcharts; reflect the actual practice; involve users in writing; minimise the paperwork; use IT (document management systems).

Document	Question it answers
Quality manual	What is our policy?
Procedure	Who does what and when?
Work instruction	How is it done?
Record	What was done / what was the result?

5. Quality Auditing

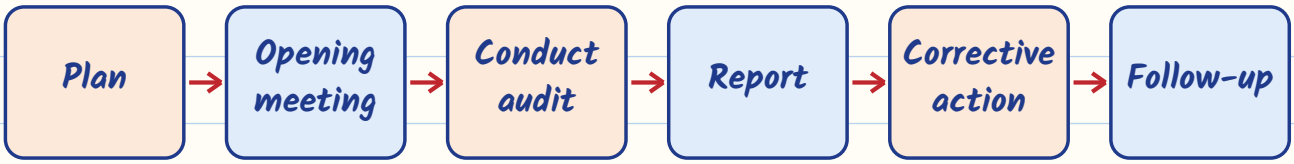
Audit: a systematic, independent and documented process for obtaining audit evidence and evaluating it objectively to determine the extent to which the **audit criteria** are fulfilled. (ISO 19011; ISO 9000:2005)

- Purpose: verify conformance with the standard and procedures, effectiveness of the system, identify improvement areas, prepare for certification, give confidence.

Types of audit

- **First party (internal) audit:** organisation audits itself, by its own trained auditors. Required by ISO 9001 at planned intervals.
- **Second party audit:** customer audits its supplier (vendor audit).
- **Third party audit:** by an independent external body (certification, surveillance audits).
- Also: system audit, process audit, **product audit**, compliance audit.

The audit process (phases)



1. **Audit planning:** scope, criteria, schedule, team (lead auditor), checklist preparation, notice to the auditee.
2. **Opening meeting** - purpose, scope, programme, and the method.
3. **Execution:** collecting evidence by interviewing, observation and examining documents and records; compare with criteria; record findings.
4. **Findings: Conformity, Non-conformity (major / minor),** observation / opportunity for improvement (OFI).
5. **Closing meeting** - present findings; agree on corrective action

timelines.

6. **Audit report** – prepared and distributed by the lead auditor.
7. **Corrective action and follow-up** by the auditee; verification of the effectiveness; closure.

- **Major NC:** absence or total breakdown of a system element/requirement; **Minor NC:** single lapse or isolated failure.

Qualities of an auditor

- Independent and objective, honest and ethical; knowledgeable of the standard and the process; good listener and communicator; observant; tactful and diplomatic; analytical; fair in reporting – "audit the system, not the person".

Audit checklist and records

- Prepared in advance with questions linked to the clause; kept as a record; helps cover all points.

Benefits of audit

- Early detection of the problems; compliance; employee awareness; basis for management review.

6. QS 9000

QS-9000 was the quality system requirement for suppliers to the automotive industry, developed by the **Chrysler, Ford and General Motors** Supplier Quality Requirements Task Force (1994). It was **ISO 9001:1994 + automotive specific requirements** (replaced later by **ISO/TS 16949** in 2002 and IATF 16949:2016).

- Aim: to **harmonise** the different supplier quality requirements of the Big Three and reduce duplication (one system for all three customers).
- Emphasis: **defect prevention,** reduction of variation and waste in the supply chain, **continuous improvement.**

Structure of QS 9000

- **Section I:** ISO 9001 requirements + additional automotive requirements (20 elements).
- **Section II: Sector specific requirements** (e.g., tooling and equipment suppliers; Ford, GM, Chrysler specific).
- **Section III: Customer specific requirements.**

Reference manuals (core tools)

- **APQP** - Advanced Product Quality Planning and Control Plan
- **FMEA** - Failure Mode and Effects Analysis
- **MSA** - Measurement System Analysis
- **SPC** - Statistical Process Control
- **PPAP** - Production Part Approval Process



Additional requirements (examples)

- Continuous improvement (quality and productivity), manufacturing capability, **process capability (Cpk >= 1.33)**, preventive maintenance, **tooling management, production part approval, design for manufacture, customer satisfaction,** supplier development, **business planning and benchmarking, bench marking, employees' empowerment.**

ISO 9001	QS 9000
General for any industry	Automotive supplier specific
System assurance	System + prevention + continuous improvement
Optional tools	Mandatory core tools (APQP, FMEA, SPC...)
Certified by registrar	Certified by an accredited registrar, recognised by the Big 3

7. ISO 14000 - Environmental Management System (EMS)

Concept

ISO 14000 is a family of standards on **environmental management** by ISO TC 207 (first issued 1996; **ISO 14001:2015** is the requirements standard). It helps an organisation **minimise the harmful effects on the environment** caused by its activities and achieve continual improvement of environmental performance.

- Based on the **PDCA:** Plan (environmental aspects, objectives), Do (implement), Check (monitor), Act (review and improve).
- Applies to all types and sizes of organisations; not a performance standard - system standard; **certifiable (14001 only).**
- Environmental aspect = element of activities that can interact with the environment (emission, effluent, waste, energy use); **Environmental impact** = the change in the environment.

The ISO 14000 family

Standard	Subject
ISO 14001	EMS requirements with guidance
ISO 14004	EMS guidelines
ISO 14010-14012 / 19011	Environmental auditing
ISO 14020	Environmental labels and declarations
ISO 14030	Environmental performance evaluation
ISO 14040	Life cycle assessment (LCA)

Requirements of ISO 14001 (core elements)

1. **Environmental policy** by top management - commitment to compliance, prevention of pollution and continual improvement.
2. **Planning:** environmental aspects and significant impacts; legal and other requirements; objectives, targets and programmes.

3. **Implementation and operation:** resources, roles and responsibilities, competence and training, communication, documentation, document control, operational control, emergency preparedness and response.
4. **Checking:** monitoring and measurement, evaluation of compliance, nonconformity and corrective action, control of records, **internal audit.**
5. **Management review** and continual improvement.



Benefits of ISO 14000

- Legal compliance and lower risk of fines; reduced consumption of energy, water and raw materials; waste reduction - cost saving.
- Better public image and community relations; marketing advantage and **green** credentials; access to global market and supply chains requiring EMS.
- Easier approval from regulators; lower insurance; employee morale; improved environmental performance; sustainability.

Implementation steps

- Commitment -> initial environmental review -> policy -> aspects and legal register -> objectives/programme -> documentation -> training -> operate -> internal audit -> management review -> certification.

ISO 9001 vs ISO 14001

Point	ISO 9001	ISO 14001
Focus	Customer / product quality	Environment / pollution prevention

Aim	Customer satisfaction	Environmental performance, compliance
Interested party	Customer	Society, regulators, community

8. Case Studies

Case 1 - ISO 9001 in a small engineering unit

- Problem: high rejection (8%), delivery delays, customer audits repeated.
- Action: gap analysis, SOPs, internal audits, training of 40 employees, supplier rating, corrective action system.
- Result: certified in 9 months; rejections down to 2%; delivery on time rose from 70% to 95%; orders increased with export customers.

Case 2 - QS 9000 at an automotive supplier (India)

- A component maker for Ford and GM implemented APQP, FMEA, SPC and PPAP; Cpk improved from 0.9 to 1.5; ppm fell from 4000 to under 100; won the Customer's supplier award.

Case 3 - ISO 14001 at a cement/paper plant

- Identified dust, energy and waste as significant aspects; installed bag filters, recycled waste heat and water; reduction in energy cost by 12%, zero legal violations; improved relations with the nearby villagers.

Case 4 - Service sector: hospital with ISO 9001

- Documented patient processes, checklists, audit system; lowered waiting time and infection rate; better patient feedback and NABH accreditation readiness.

Case 5 - Integrated management system (IMS)

- Companies combine ISO 9001 + 14001 + 45001 into one system to save documents and audits.

> How to write a case answer: Background -> Problem -> System applied -> Result -> Lesson.

Important Exam Questions (Unit V)

Short answer

- Why is ISO 9000 needed? List the 8 principles of ISO 9000:2000.
- Mention the four levels of documentation.
- Differentiate between internal and external audits.
- What are the core manuals of QS 9000?
- State any four benefits of ISO 14000.

Long answer

1. Explain the elements of ISO 9001:2000 quality system.
2. Explain the steps for implementing a quality system. How is certification obtained?
3. Explain the documentation of the quality system with a pyramid.
4. Describe the quality audit process and the types of audit.
5. Explain QS 9000 and its requirements.
6. Explain the ISO 14000 concept, requirements and benefits.
7. Discuss a case study on implementation of ISO 9000 / ISO 14000.